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EST II — MATH LEVEL 1 • UNIT 0.01

EST II Math Level 1 Diagnostic & Strategy Drill

ANSWER KEY & SOLUTIONS

Core Subject Readiness • Official EST II Curriculum

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 C	Q3 B	Q4 C	Q5 D
Q6 D	Q7 B	Q8 D	Q9 A	Q10 A
Q11 C	Q12 B	Q13 B	Q14 C	Q15 C
Q16 C	Q17 C	Q18 C	Q19 B	Q20 B
Q21 B	Q22 D	Q23 D	Q24 B	Q25 C

EST II Math Level 1 Diagnostic & Strategy Drill Practice Exercises

1 MCQ

Consider the function $g(x) = 2x + 3$. What is the value of $g(4)$?

A 7

B 8

C 11 ✓

D 14

WORKED SOLUTION & VERIFICATION

To find the value of $g(4)$, we substitute $x = 4$ into the function $g(x) = 2x + 3$.

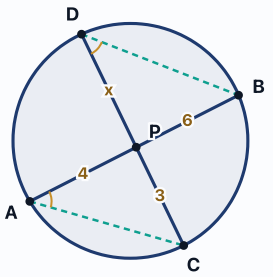
$g(4) = 2(4) + 3$

$g(4) = 8 + 3$

$g(4) = 11$ Therefore, the value of $g(4)$ is 11.

2 MCQ

In the circle shown below, chords AB and CD intersect at point P. If $PA = 4$, $PB = 6$, and $PC = 3$, what is the length of segment $\overline{PD}$?



A 5

B 7

C 8 ✓

D 9

WORKED SOLUTION & VERIFICATION

By the Intersecting Chords Theorem (Power of a Point Theorem), when two chords AB and CD intersect at an interior point P, the products of their segments are equal:

$$PA \cdot PB = PC \cdot PD$$

Substitute the given values into the equation:

$$(4)(6) = (3)(PD)$$

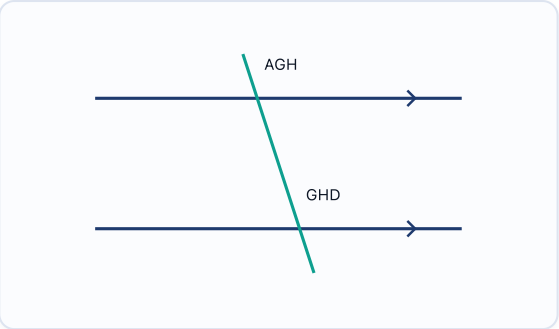
$$24 = 3 \cdot PD$$

$$PD = \frac{24}{3} = 8$$

Common error analysis:
- Adding the segment lengths instead of multiplying ($PA + PB = PC + PD \implies 4 + 6 = 3 + PD \implies PD = 7$) yields 7, which fails because the segments satisfy a multiplicative relationship derived from similar triangles ($\triangle PAD \sim \triangle PCB$).

3 MCQ

In the diagram below, line AB is parallel to line CD. Transversal line EF intersects AB at point G and CD at point H. If the measure of angle AGH is 50°, what is the measure of angle GHD?



A 50°

B 130° ✓

C 40°

D 90°

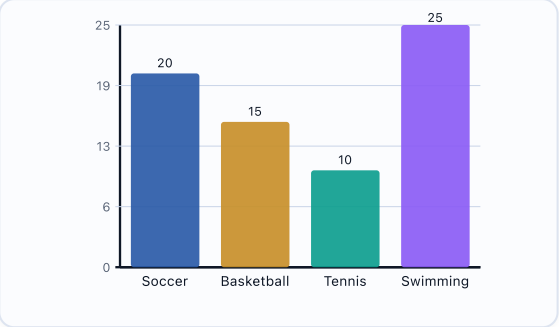
WORKED SOLUTION & VERIFICATION

Since line AB is parallel to line CD, $\angle AGH$ and $\angle CHG$ are alternate interior angles, so $m\angle CHG = m\angle AGH = 50^\circ$. Since $\angle CHG$ and $\angle GHD$ form a linear pair along line CD, they are supplementary:

$$m\angle GHD = 180^\circ - 50^\circ = 130^\circ.$$

4 MCQ

The bar chart below shows the number of students who chose different sports as their favorite.
What is the total number of students surveyed?



A 50

B 60

C 70 ✓

D 80

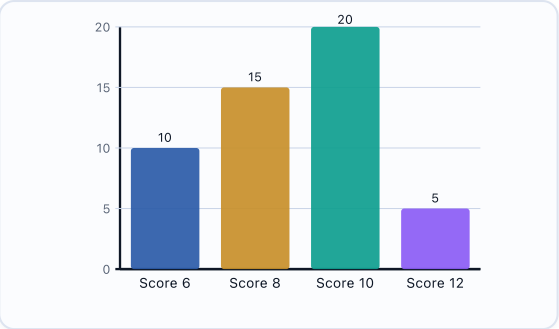
WORKED SOLUTION & VERIFICATION

To find the total number of students surveyed, we need to sum the number of students for each sport.
From the bar chart:
- Soccer: 20 students
- Basketball: 15 students
- Tennis: 10 students
- Swimming: 25 students

Total students = 20 + 15 + 10 + 25 = 70
Therefore, the total number of students surveyed is 70.

5 MCQ

The bar chart below shows the number of students who achieved different scores on a recent quiz. What is the total number of students who scored 8 or 10?



A 25

B 30

C 40

D **35** ✓

WORKED SOLUTION & VERIFICATION

The bar chart shows that 15 students scored 8 and 20 students scored 10. Therefore, the total number of students who scored 8 or 10 is $15 + 20 = 35$.

6 MCQ

For $\mathbf{a} = \langle 3, 4 \rangle$ and $\mathbf{b} = \langle 4, -3 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 12, -12 \rangle$

B $\langle 9, 0 \rangle$

C $\langle -1, 7 \rangle$

D $\langle 7, 1 \rangle$ ✓

WORKED SOLUTION & VERIFICATION

$\mathbf{a} + \mathbf{b} = \langle 3 + (4), 4 + (-3) \rangle = \langle 7, 1 \rangle$.

7 MCQ

Given the function $f(x) = 2x + 3$. What is the value of $f^{-1}(7)$?

A 1

B **2** ✓

C 3

D 4

WORKED SOLUTION & VERIFICATION

Let $y = f(x)$. Then $y = 2x + 3$. To find the inverse function $f^{-1}(y)$, we swap x and y and solve for y.
So, $x = 2y + 3$.
Subtract 3 from both sides: $x - 3 = 2y$.
Divide by 2: $y = \frac{x-3}{2}$.
Thus, $f^{-1}(x) = \frac{x-3}{2}$.
We need to find $f^{-1}(7)$. Substitute $x = 7$ into the inverse function:
$f^{-1}(7) = \frac{7-3}{2} = \frac{4}{2} = 2$.

8 MCQ

For $\mathbf{a} = \langle 4, 7 \rangle$ and $\mathbf{b} = \langle 3, -4 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 9, 2 \rangle$

B $\langle 12, -28 \rangle$

C $\langle 1, 11 \rangle$

D $\langle 7, 3 \rangle$ ✓

WORKED SOLUTION & VERIFICATION

$\mathbf{a} + \mathbf{b} = \langle 4 + (3), 7 + (-4) \rangle = \langle 7, 3 \rangle$.

9 MCQ

For $\mathbf{a} = \langle 5, 10 \rangle$ and $\mathbf{b} = \langle 2, -5 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 7, 5 \rangle$ ✓

B $\langle 10, -50 \rangle$

C $\langle 9, 4 \rangle$

D $\langle 3, 15 \rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{a} + \mathbf{b} = \langle 5 + (2), 10 + (-5) \rangle = \langle 7, 5 \rangle$.

10 MCQ

For $\mathbf{a} = \langle 6, 13 \rangle$ and $\mathbf{b} = \langle 1, -6 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 7, 7 \rangle$ ✓

B $\langle 5, 19 \rangle$

C $\langle 9, 6 \rangle$

D $\langle 6, -78 \rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{a} + \mathbf{b} = \langle 6 + (1), 13 + (-6) \rangle = \langle 7, 7 \rangle$.

11 MCQ

For $\mathbf{a} = \langle 7, 16 \rangle$ and $\mathbf{b} = \langle 0, -7 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 7, 23 \rangle$

B $\langle 9, 8 \rangle$

C $\langle 7, 9 \rangle$ ✓

D $\langle 0, -112 \rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{a} + \mathbf{b} = \langle 7 + (0), 16 + (-7) \rangle = \langle 7, 9 \rangle$.

12 MCQ

For $\mathbf{a} = \langle 8, 19 \rangle$ and $\mathbf{b} = \langle -1, -8 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 9, 27 \rangle$

B $\langle 7, 11 \rangle$ ✓

C $\langle -8, -152 \rangle$

D $\langle 9, 10 \rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{a} + \mathbf{b} = \langle 8 + (-1), 19 + (-8) \rangle = \langle 7, 11 \rangle$.

13 MCQ

For $\mathbf{a} = \langle 9, 22 \rangle$ and $\mathbf{b} = \langle -2, -9 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 9, 12 \rangle$

B $\langle 7, 13 \rangle$ ✓

C $\langle 11, 31 \rangle$

D $\langle -18, -198 \rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{a} + \mathbf{b} = \langle 9 + (-2), 22 + (-9) \rangle = \langle 7, 13 \rangle$.

14 MCQ

For $\mathbf{a} = \langle 10, 25 \rangle$ and $\mathbf{b} = \langle -3, -10 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle 9, 14 \rangle$

B $\langle -30, -250 \rangle$

C $\langle 7, 15 \rangle$ ✓

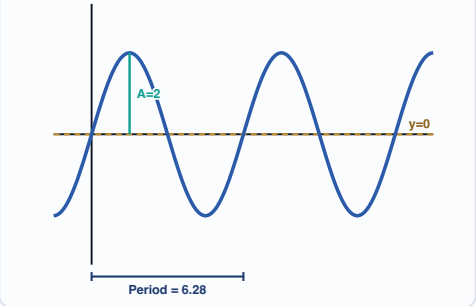
D $\langle 13, 35 \rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{a} + \mathbf{b} = \langle 10 + (-3), 25 + (-10) \rangle = \langle 7, 15 \rangle$.

15 MCQ

For $\mathbf{a} = \langle 11, 28 \rangle$ and $\mathbf{b} = \langle -4, -11 \rangle$, what is $\mathbf{a} + \mathbf{b}$?



A $\langle 9, 16 \rangle$

B $\langle -44, -308 \rangle$

C $\langle 7, 17 \rangle$ ✓

D $\langle 15, 39 \rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{a} + \mathbf{b} = \langle 11 + (-4), 28 + (-11) \rangle = \langle 7, 17 \rangle$.

16 MCQ

For $\mathbf{a} = \langle 12, 31 \rangle$ and $\mathbf{b} = \langle -5, -12 \rangle$, what is $\mathbf{a} + \mathbf{b}$?

A $\langle -60, -372 \rangle$

B $\langle 9, 18 \rangle$

C $\langle 7, 19 \rangle$ ✓

D $\langle 17, 43 \rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{a} + \mathbf{b} = \langle 12 + (-5), 31 + (-12) \rangle = \langle 7, 19 \rangle$.

17 MCQ

Find the angle θ between $\mathbf{u} = \langle 1, 1\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 3 \rangle$.

A 90°

B 60°

C 30° ✓

D 45°

WORKED SOLUTION & VERIFICATION

$\mathbf{u} \cdot \mathbf{v} = 0 + (k+2)\sqrt{3}$. $\|\mathbf{u}\| = 2k$, $\|\mathbf{v}\| = k+2$. $\cos\theta = \frac{\sqrt{3}}{2} \implies \theta = 30^\circ$.

18 MCQ

Calculate the angle θ between $\mathbf{u} = \langle 2, 2\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 4 \rangle$.

A 90°

B 60°

C 30° ✓

D 45°

WORKED SOLUTION & VERIFICATION

$\mathbf{u} \cdot \mathbf{v} = 0 + 2(k+2)\sqrt{3}$. $\|\mathbf{u}\| = 2k$, $\|\mathbf{v}\| = k+2$. $\cos\theta = \frac{\sqrt{3}}{2} \implies \theta = 30^\circ$.

19 MCQ

Compute the angle θ between $\mathbf{u} = \langle 3, 3\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 5 \rangle$.

A 45°

B 30° ✓

C 90°

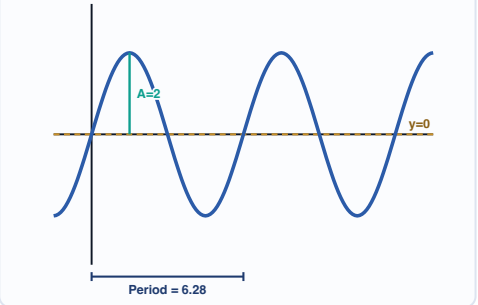
D 60°

WORKED SOLUTION & VERIFICATION

$\mathbf{u} \cdot \mathbf{v} = 0 + 3(k+2)\sqrt{3}$. $\|\mathbf{u}\| = 2k$, $\|\mathbf{v}\| = k+2$. $\cos\theta = \frac{\sqrt{3}}{2} \implies \theta = 30^\circ$.

20 MCQ

Deduce the angle θ between $\mathbf{u} = \langle 4, 4\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 6 \rangle$.



A 60°

B 30° ✓

C 90°

D 45°

WORKED SOLUTION & VERIFICATION

$\mathbf{u} \cdot \mathbf{v} = 0 + 4(k+2)\sqrt{3}$. $\|\mathbf{u}\| = 2k$, $\|\mathbf{v}\| = k+2$. $\cos\theta = \frac{\sqrt{3}}{2} \implies \theta = 30^\circ$.

21 MCQ

Obtain the angle θ between $\mathbf{u} = \langle 5, 5\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 7 \rangle$.

A 60°

B 30° ✓

C 90°

D 45°

WORKED SOLUTION & VERIFICATION

$\mathbf{u} \cdot \mathbf{v} = 0 + 5(k+2)\sqrt{3}$. $\|\mathbf{u}\| = 2k$, $\|\mathbf{v}\| = k+2$. $\cos\theta = \frac{\sqrt{3}}{2} \implies \theta = 30^\circ$.

22 MCQ

Identify the angle θ between $\mathbf{u} = \langle 6, 6\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 8 \rangle$.

A 45°

B 90°

C 60°

D 30° ✓

WORKED SOLUTION & VERIFICATION

$\mathbf{u} \cdot \mathbf{v} = 0 + 6(k+2)\sqrt{3}$. $\|\mathbf{u}\| = 2k$, $\|\mathbf{v}\| = k+2$. $\cos\theta = \frac{\sqrt{3}}{2} \implies \theta = 30^\circ$.

23 MCQ

Establish the angle θ between $\mathbf{u} = \langle 7, 7\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 9 \rangle$.

A 90°

B 45°

C 60°

D 30° ✓

WORKED SOLUTION & VERIFICATION

$\mathbf{u} \cdot \mathbf{v} = 0 + 7(k+2)\sqrt{3}$. $\|\mathbf{u}\| = 2k$, $\|\mathbf{v}\| = k+2$. $\cos\theta = \frac{\sqrt{3}}{2} \implies \theta = 30^\circ$.

24 MCQ

Determine the angle θ between $\mathbf{u} = \langle 8, 8\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 10 \rangle$.

A 60°

B 30° ✓

C 90°

D 45°

WORKED SOLUTION & VERIFICATION

$\mathbf{u} \cdot \mathbf{v} = 0 + 8(k+2)\sqrt{3}$. $\|\mathbf{u}\| = 2k$, $\|\mathbf{v}\| = k+2$. $\cos\theta = \frac{\sqrt{3}}{2} \implies \theta = 30^\circ$.

25 MCQ

Calculate the angle θ between $\mathbf{u} = \langle 9, 9\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 11 \rangle$.

A 60°

B 90°

C 30° ✓

D 45°

WORKED SOLUTION & VERIFICATION

$\mathbf{u} \cdot \mathbf{v} = 0 + 9(k+2)\sqrt{3}$. $\|\mathbf{u}\| = 2k$, $\|\mathbf{v}\| = k+2$. $\cos\theta = \frac{\sqrt{3}}{2} \implies \theta = 30^\circ$.